

KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
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SMART LIBRARY BOOK RECOMMENDER

Project ID : PRJ029

Team ID : T029

Team Name : TechNexora

Team GitHub Repository : https://github.com/Gani905911/Gen-Ai-

Website Live Link : https://gen-sage-seven.vercel.app

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K. Charan Sai	99230040353	https://github.com/charan1821/GENAI-ASSESMENT
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Project Trainer / Guide

Mr. SHIRWIN PRINCE I

I. INTRODUCTION

The Smart Library Book Recommender is an AI-powered web application that helps users discover and explore books based on their interests and preferences. It uses Artificial Intelligence, Semantic Search, and Retrieval-Augmented Generation (RAG) to provide personalized book recommendations.

The application integrates the Google Books API to retrieve book details such as titles, authors, descriptions, and cover images. Users can search for books by title, author, category, or natural-language queries.

The system uses Sentence Transformers and FAISS for semantic search. Firebase Authentication provides secure login, while Cloud Firestore stores user profiles, saved books, ratings, and reading history.

Users can also read available books online, access PDFs where permitted, and track their reading progress by saving completed pages.

The main objective of this project is to make book discovery easier and provide a personalized digital reading experience through modern AI technologies.

II. PROBLEM STATEMENT & OBJECTIVES

PROBLEM : Traditional library systems often make it difficult for users to find books that match their interests and reading preferences. Searching through large collections manually takes time and does not provide personalized recommendations. Users may also face difficulties in managing their reading lists and tracking completed pages.

To overcome these challenges, the Smart Library Book Recommender provides an AI-powered web application that helps users discover relevant books, receive personalized recommendations, manage their digital library, and track their reading progress.

OBJECTIVES :

- To develop a user-friendly web application for discovering books.
- To provide personalized book recommendations using AI and Semantic Search.
- To integrate Google Books API for retrieving book information.
- To implement RAG and FAISS for intelligent book retrieval.
- To provide secure user authentication using Firebase.
- To allow users to save books and manage their personal library.
- To support online reading and PDF access where available.
- To track completed pages and save reading progress.
- To improve the overall book discovery and reading experience.

III. SYSTEM ARCHITECTURE

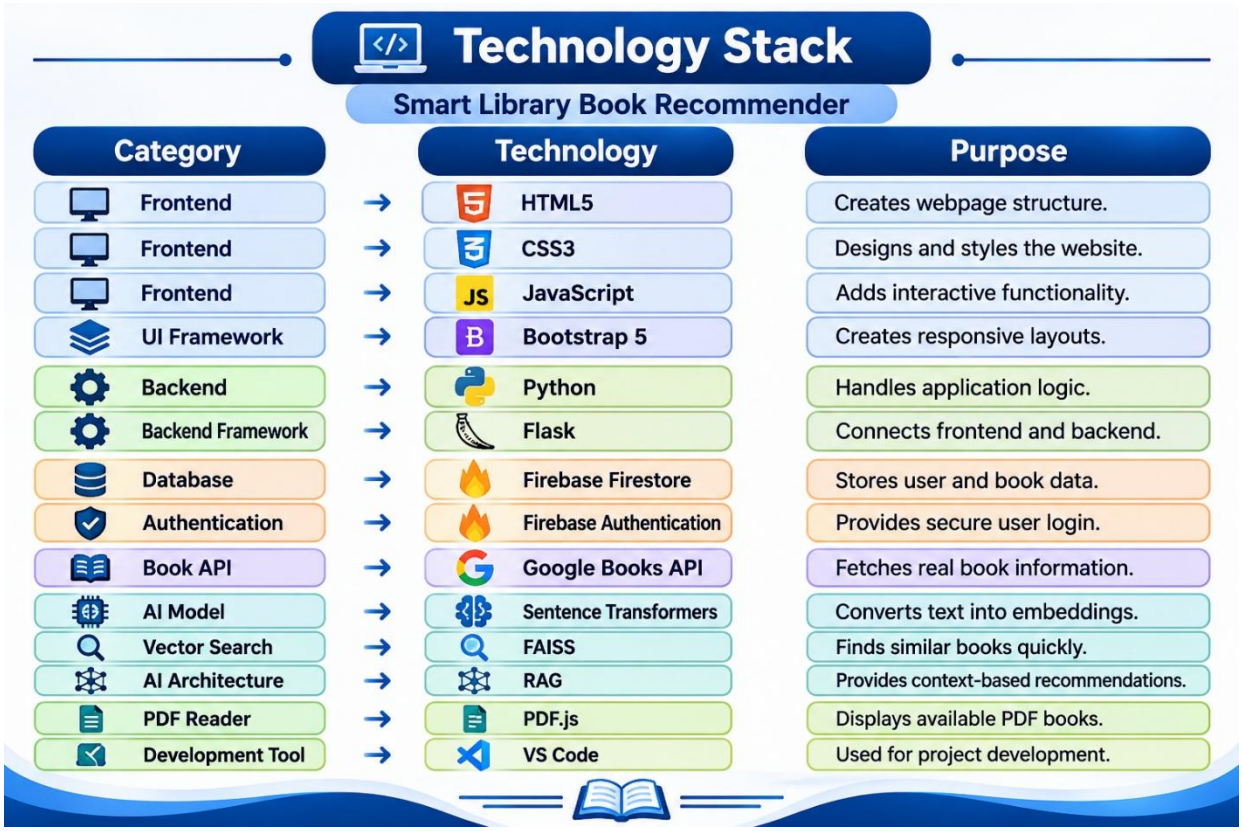


The Smart Library Book Recommender uses a Retrieval-Augmented Generation (RAG) architecture to provide relevant and personalized book recommendations. It combines Google Books API, Sentence Transformers, FAISS, and Firebase to retrieve and rank books based on user queries.

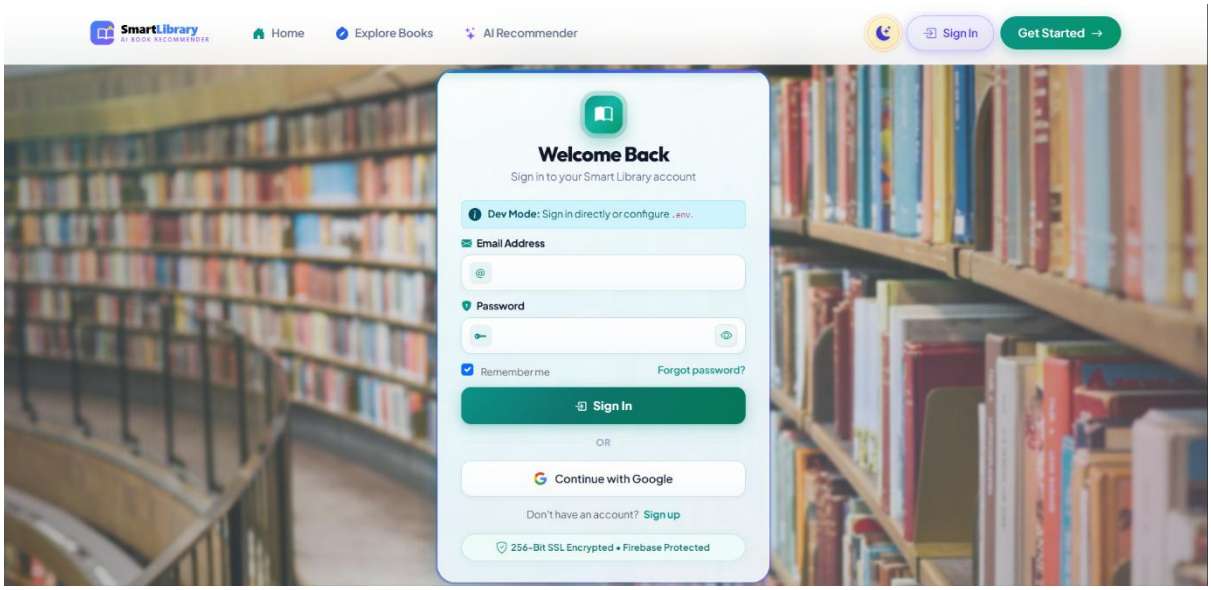
Working Process

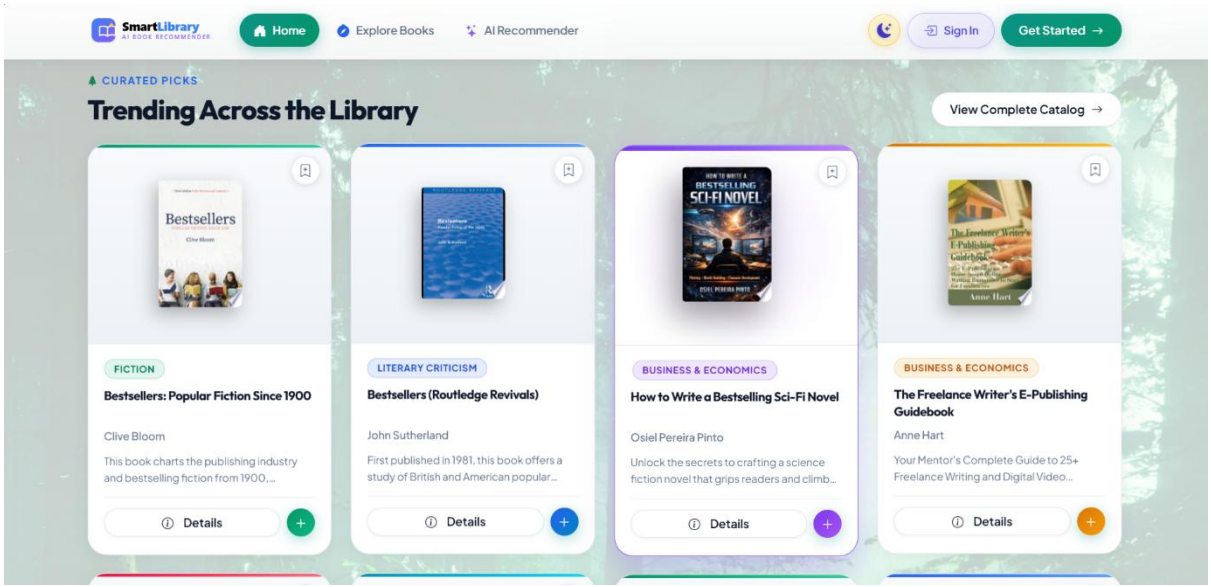
1. **User Query:** The user enters a query describing their interests, such as books about Machine Learning.
2. **Query Processing:** The system processes the query and generates numerical embeddings using Sentence Transformers.
3. **Book Retrieval:** The Google Books API retrieves relevant book information, including titles, authors, and descriptions.
4. **Vector Embedding:** Book information is converted into vector embeddings using the Sentence Transformer model.
5. **Semantic Search:** FAISS compares the query embedding with book embeddings to identify the most relevant books.
6. **Recommendation Ranking:** The system ranks the retrieved books based on semantic similarity, subject relevance, and user preferences.
7. **Response Generation:** The RAG component prepares recommendations with relevant explanations based on the retrieved book information.
8. **User Interaction:** Recommended books are displayed on the website, where users can explore, save, and read available books.

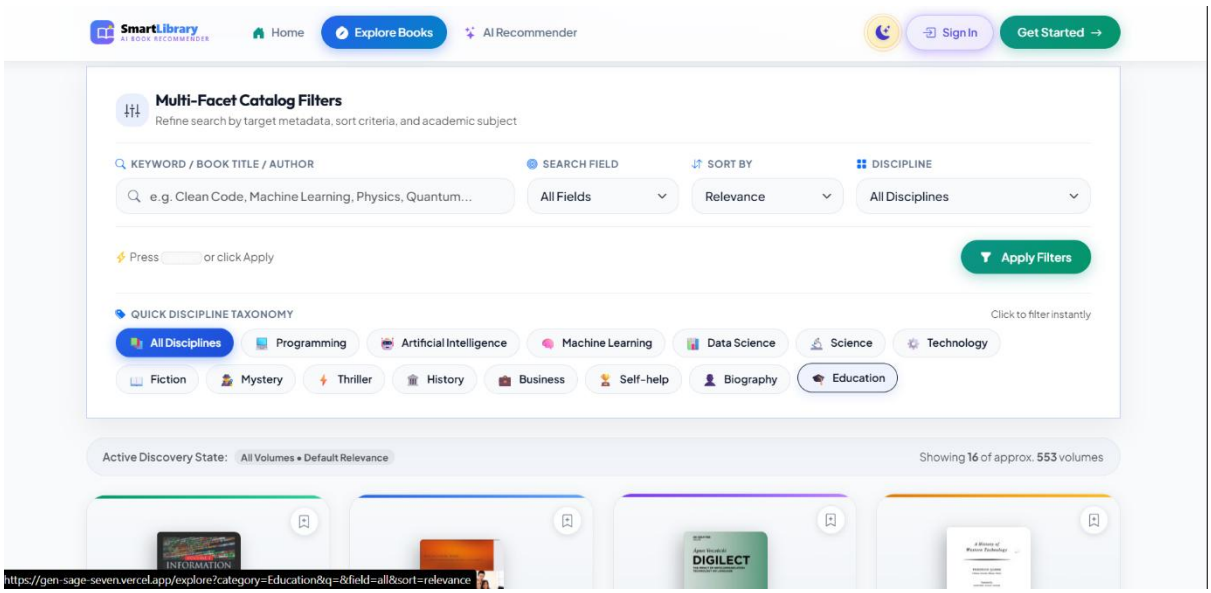
IV. TECH STACK

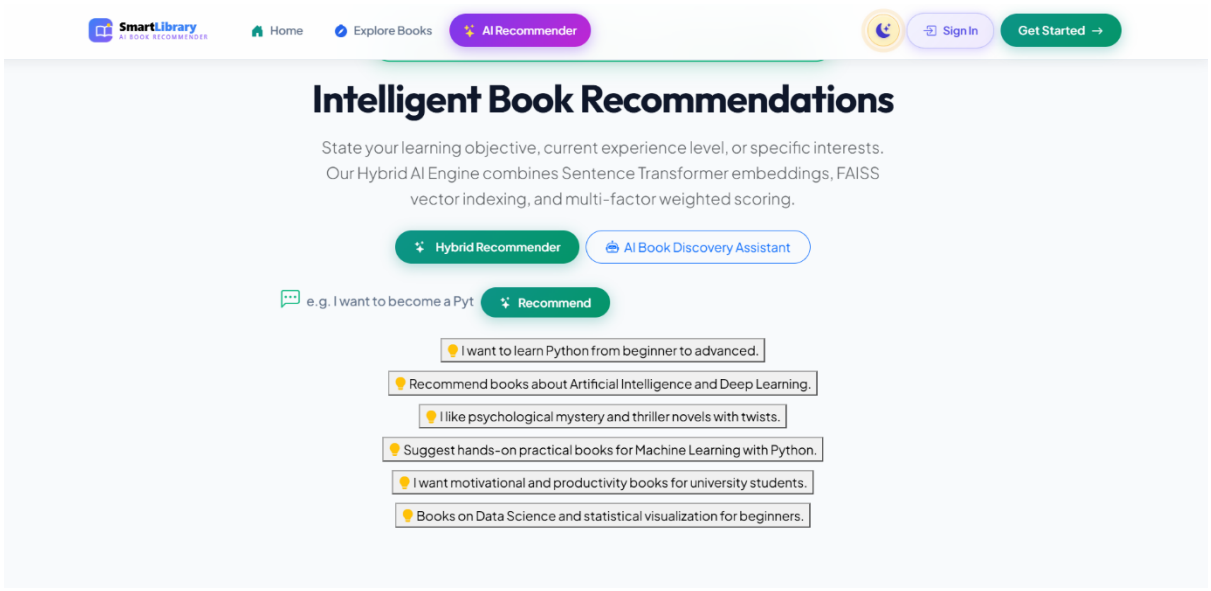


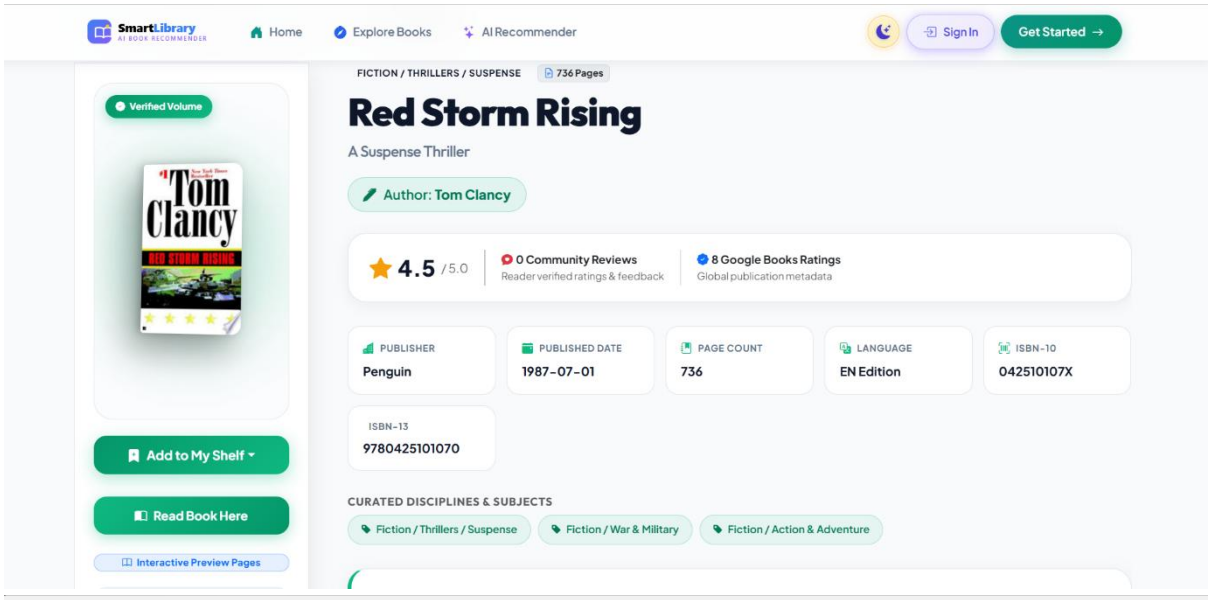
V. IMPLEMENTATION DETAILS

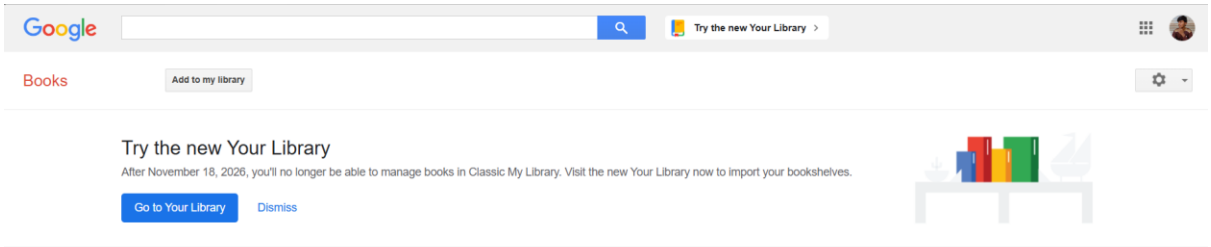



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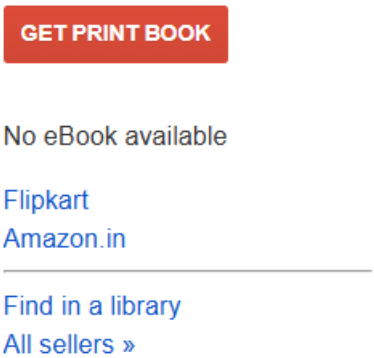

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About the author (1987)
A little more than thirty years ago Tom Clancy was a Maryland insurance broker with a passion for naval history. Years before, he had been an English major at Baltimore's Loyola College and had always dreamed of writing a novel. His first effort, The Hunt for Red October, sold briskly as a result of rave reviews, then catapulted onto the New York Times bestseller list after President Reagan pronounced it "the perfect yarn." From that day forward, Clancy established himself as an


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VI. RESULTS AND TESTING

6.1 Results

The Smart Library Book Recommender successfully provides a user-friendly platform for discovering and managing books. The application integrates AI-based semantic search and RAG to recommend relevant books based on user interests.

The system retrieves real book information through the Google Books API and allows users to save books, manage their reading lists, and track reading progress. Firebase provides user authentication and cloud data storage.

The application also supports online reading and PDF access where available, providing a convenient digital reading experience.

6.2 Testing

The following testing activities are used to verify the application's functionality:

Test Case	Expected Result
User Registration	New users can create accounts.
User Login	Registered users can log in securely.
Book Search	Relevant book details are displayed.
Google Books API	Real book information is retrieved.
AI Recommendations	Relevant books are recommended.
Semantic Search	Similar books are identified.
My Library	Users can save and manage books.
Reading Progress	Completed pages are saved.
Online Reader	Available books can be opened.
Firebase Database	User data is stored and retrieved.

6.3 Overall Outcome

The project combines AI technologies and modern web development tools to create an intelligent book discovery platform. It makes searching, recommending, managing, and reading books easier and more convenient for users.

VII. CONCLUSION & FUTURE SCOPE

7.1 CONCLUSION

The Smart Library Book Recommender provides an intelligent and user-friendly platform for discovering and managing books. By integrating AI, Semantic Search, RAG, Google Books API, and Firebase, the system offers personalized book recommendations and an improved digital reading experience. It also helps users manage their libraries and track reading progress efficiently.

7.2 FUTURE SCOPE

- Improve recommendation accuracy using advanced AI models.
- Support more digital book formats.
- Add voice-based book search.
- Provide multilingual book recommendations.
- Introduce personalized reading suggestions.
- Add reading statistics and progress reports.
- Improve the AI assistant with more conversational features.